

Thermodynamic State Vector Interface: Enforcing First and Second Law Invariants in the Web of Life Planetary Simulation Engine

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Abstract

Complex global systems ecology simulations require strict adherence to physical conservation laws to prevent unphysical energy generation or entropy destruction over long temporal horizons. Sprint 011 establishes the **Thermodynamic State Vector Interface** (`src/thermodynamics/types.ts`) within the Web of Life simulation engine. This module provides rigorous type-safe contracts and functional monad state transformations (`ThermodynamicStateMonad`) that enforce the First Law of Thermodynamics (energy and mass conservation) and the Second Law of Thermodynamics (non-negative internal entropy generation, $\dot{S}_{\text{gen}} \geq 0$). Furthermore, we formalize the Gouy-Stodola Theorem to track real-time exergy destruction ($\dot{I} = T_0 \dot{S}_{\text{gen}}$) across all biogeochemical and planetary pods relative to a standard ambient dead-state temperature ($T_0 = 288.15 \text{ K}$). All source code and verification suites are available in the official repository: <https://github.com/pascalranoroarijaona/WebOfLife>.

1 Introduction & Systems Ecology Motivation

Planetary-scale ecological models often suffer from numerical drift or unconstrained energy transformations, leading to unphysical behaviors such as spontaneous energy creation or violations of the Clausius inequality. To maintain thermodynamic credibility, the Web of Life framework models the global Earth system as an open thermodynamic control volume (Ω) interacting with solar radiation inputs and deep space thermal sinks.

Sprint 011 introduces the mathematical foundation and type definitions necessary to rigorously track:

1. Internal energy accumulation and boundary heat/mass exchanges.
2. Irreversible internal entropy generation rates ($\dot{S}_{\text{gen}}$).
3. Exergy destruction ($\dot{I}$), quantifying lost work potential and thermodynamic dissipation across ecological trophic levels and biogeochemical cycles (Carbon, Nitrogen, Phosphorus, Water).

2 Governing Thermodynamic Equations

2.1 First Law: Energy Conservation

The net rate of change of internal energy within the system control volume is governed by the First Law of Thermodynamics:

$$\frac{dE_{\text{sys}}}{dt} = \dot{Q}_{\text{net}} - \dot{W}_{\text{net}} + \sum_{\text{in}} \dot{m}_{\text{in}} h_{\text{in}} - \sum_{\text{out}} \dot{m}_{\text{out}} h_{\text{out}} \quad (1)$$

where $\dot{Q}_{\text{net}}$ is the net heat flux crossing the boundary, $\dot{W}_{\text{net}}$ is the net work done by the system, and $h_{\text{in}}/h_{\text{out}}$ represent specific enthalpies associated with boundary mass fluxes.

2.2 Second Law & Exergy Destruction

The entropy balance across the system boundary is expressed as:

$$\frac{dS_{\text{sys}}}{dt} = \sum_k \frac{\dot{Q}_k}{T_k} + \sum_{\text{in}} \dot{m}_{\text{in}} s_{\text{in}} - \sum_{\text{out}} \dot{m}_{\text{out}} s_{\text{out}} + \dot{S}_{\text{gen}} \quad (2)$$

Per the **Second Law of Thermodynamics** (Entropy Generation Postulate), internal irreversibilities require:

$$\dot{S}_{\text{gen}} \geq 0 \quad (\text{Strict Invariant}) \quad (3)$$

Using the **Gouy-Stodola Theorem**, the exergy destruction rate ($\dot{I}$) is linked directly to entropy generation via the reference dead-state temperature ($T_0 = 288.15 \text{ K}$):

$$\dot{I} = T_0 \dot{S}_{\text{gen}} \geq 0 \quad (4)$$

3 Implementation: Type Contracts & Monad State Propagation

The interface contracts are implemented in TypeScript to enforce compile-time safety and run-time invariant checks.

3.1 Core Type Definitions (src/thermodynamics/types.ts)

```
export const STANDARD_AMBIENT_TEMPERATURE_K = 288.15;

export interface ThermalFluxVector {
  solarInbound: number;      // W (>= 0)
  thermalOutbound: number;   // W
  sensibleHeatFlux: number;  // W
}

export interface MassFluxVector {
  massInflowRate: number;     // kg/s
  massOutflowRate: number;    // kg/s
  specificEnthalpyIn: number; // J/kg
  specificEnthalpyOut: number; // J/kg
  specificEntropyIn: number;  // J/(kg*K)
  specificEntropyOut: number; // J/(kg*K)
}

export interface IThermodynamicStateVector {
  timestamp: number;
  internalEnergy: number;      // Joules
  systemEntropy: number;      // J/K
  entropyGenerationRate: number; // J/(K*s) [Must be >= 0]
  exergyDestructionRate: number; // Watts [I = T_0 * S_gen]
  thermalFluxes: ThermalFluxVector;
  massFluxes: MassFluxVector;
  referenceTemperature: number; // Kelvin
}
```

3.2 Functional State Monad

State transitions are wrapped in the `ThermodynamicStateMonad`, ensuring that every simulation step validates Second Law compliance prior to committing state updates:

```
export function advanceThermodynamicState(
  currentState: IThermodynamicStateVector,
  deltaEnergy: number,
  entropyGenRate: number,
  dt: number
): IThermodynamicStateVector {
  if (entropyGenRate < 0) {
    throw new Error('Second Law Violation: S_gen (${entropyGenRate}) < 0.');
```



```
const T_0 = currentState.referenceTemperature ?? STANDARD_AMBIENT_TEMPERATURE_K;
const exergyDestructionRate = T_0 * entropyGenRate;

return {
  timestamp: currentState.timestamp + dt,
  internalEnergy: currentState.internalEnergy + deltaEnergy,
  systemEntropy: currentState.systemEntropy + (entropyGenRate * dt),
  entropyGenerationRate: entropyGenRate,
  exergyDestructionRate: exergyDestructionRate,
  thermalFluxes: { ...currentState.thermalFluxes },
  massFluxes: { ...currentState.massFluxes },
  referenceTemperature: T_0
};
}
```

4 Conclusion & Future Directions

Sprint 011 successfully establishes the thermodynamic guardrails for the Web of Life engine. By enforcing strict boundary flux tracking, mass and energy conservation, and non-negative entropy generation, the simulation engine prevents physical drift. Future sprints will integrate these thermodynamic vectors directly into nutrient cycling and trophic energy transfer efficiencies within planetary pods.

Repository Reference: <https://github.com/pascalranoroarijaona/WebOfLife>